Evidence-bank contamination and order-sensitivity experiment

Methodological design for the frozen Stage 2 DEMA pipeline

Purpose and design

A controlled sensitivity experiment was conducted to test two potential pipeline effects: (1) contamination, defined as the effect of adding low-quality evidence records to an otherwise unchanged evidence bank; and (2) order sensitivity, defined as variation in the output when identical evidence records were presented in different sequences. Indicator A.1.2, Roles are known and identified for the coordinator, was selected as the reference indicator. The purpose was not to re-estimate overall DEMA maturity, but to determine whether evidence-bank presentation affected the phase, the records used to justify that phase, or the records classified as limiting.

The complete 622-record evidence bank used by the frozen pipeline was retained. No record was selected or excluded on the basis of its quality rating or its use in an earlier justification. This full-bank design avoided circular selection and preserved the informational environment encountered by the operational scoring call. The experimental unit was one Stage 2 assessment of A.1.2 under a specified evidence ordering and contamination condition.

Experimental conditions

Component	Clean condition	Contaminated condition
Genuine evidence	All 622 frozen-bank records	The identical 622 records
Synthetic controls	None	All 155 controls
Genuine-record order	Seeded random permutation	Identical to paired clean run
Control placement	Not applicable	Inserted at seeded positions
Replications	10 permutations	10 paired treatments
Scoring target	A.1.2 only	A.1.2 only

Ten reproducible seeds (20261001–20261010) generated the clean-bank permutations. For each seed, a paired contaminated bank preserved the exact relative order of the 622 genuine records and inserted 155 controls at seeded positions. The pairing therefore held genuine evidence and its internal order constant when estimating the contamination effect. Across the experiment, 20 Stage 2 calls were executed: ten clean and ten contaminated.

Controlled pipeline parameters

The Stage 2 prompt was retrieved from Langfuse as the frozen P5 version and verified against its recorded SHA-256 hash before execution. Prompt text, model, temperature (0), country, indicator definition, output schema, maximum output allowance and validation procedure were held constant. All genuine and synthetic records were assigned neutral EV-#### identifiers. Record provenance and experimental type were retained in an analyst-only manifest and were not disclosed to the scoring model. Each call was traced, and the Langfuse client was flushed after every paired comparison.

Construction of the contamination treatment

The contamination set comprised five deliberately low-quality synthetic records for each of the 31 DEMA indicators (155 records in total). The controls represented recurring forms of superficially relevant but maturity-inadequate material: an agenda mention, presentation attendance, a generic external comment, a future possibility and a glossary reference. Five controls therefore targeted A.1.2, while the remaining 150 provided broader evidence-bank noise. The controls were not derived from the documentary corpus and were introduced solely for the sensitivity test. Their interpretation boundaries explicitly stated why they did not demonstrate implementation, institutionalization, outcomes or a maturity-relevant country condition.

Outcome measures

Outcomes were evaluated separately at score, positive-justification and limiting-evidence levels. Separating these levels was necessary because an unchanged phase can conceal changes in the records selected to support it, and a record used as a limitation performs the opposite function from a record used to justify maturity.

Metric	Definition and interpretation
Phase distribution	Frequency of each assigned phase across the ten clean permutations; measures order-related score variation.
Pairwise phase agreement	Proportion of the 45 clean-run pairs receiving the same phase; 100% denotes phase invariance to order.
Paired phase stability	Proportion of clean–contaminated pairs with identical phases; measures score stability after controls are added.
Supporting-set Jaccard	$\|S_1 \cap S_2\| / \|S_1 \cup S_2\|$ for positive evidence-ID sets; 1 denotes identical selection and 0 no overlap.
Supporting-evidence retention	|Sclean ∩ Streatment| / |Sclean|, restricted to genuine records; measures how much clean support remains after contamination.
Limiting-set Jaccard	Jaccard similarity of contrary-or-limiting evidence-ID sets; measures stability of the evidentiary qualification.
Positive contamination rate	Control citations in evidence_supporting_score divided by 1,550 control record-run exposures; measures direct uptake as maturity support.
Limiting-control rate	Control citations in contrary_or_limiting_evidence divided by 1,550 exposures; distinguishes rejection/qualification from positive contamination.
Selection frequency	Number of runs in which each genuine record was cited; identifies consistently selected and order-sensitive records.

Examples of synthetic contaminated records

Blinded ID	Control type	Evidence statement supplied to Stage 2
EV-0668	Agenda mention	A meeting agenda in Togo mentioned Roles are known & identified, for the coordinator as a discussion topic.
EV-0739	Future possibility	Participants expressed possible future interest in Roles are known & identified, for the coordinator.
EV-0672	Generic external comment	A generic international presentation referred to Roles are known & identified, for the coordinator.
EV-0741	Glossary reference	An introductory brochure included the term Roles are known & identified, for the coordinator in a glossary.
EV-0646	Presentation attendance	An introductory presentation attended in Togo mentioned Roles are known & identified, for the coordinator.

For example, the future-possibility control stated that participants expressed possible future interest in the coordinator-role indicator. Its accompanying boundary clarified that conditional interest did not document a commitment, funded activity, implementation, output or maturity-relevant condition. Similarly, the agenda and glossary controls established only that the indicator wording had been mentioned, not that a coordinator had been identified or was performing the required role.

Analytical scope and reproducibility

The experiment was analysed as a focused indicator-level case study. The ten seeded permutations constitute replications of evidence presentation, not independent samples of countries or indicators. Accordingly, results were summarized descriptively rather than generalized inferentially to the complete DEMA framework. Reproducibility was supported by fixed seeds, paired bank construction, immutable prompt versioning, temperature zero, neutral identifiers, retained input banks, analyst mappings, validated JSON outputs and call-level traces.